



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

DESIGN AND TECHNOLOGY

Paper 2 Design and Drawing

4053/2

3 hours

SPECIMEN PAPER

Additional materials

Standard, drawing equipment,
Coloured pencils,
Calculator,

Computer with AutoCAD drawing software installed,
2 x A3 drawing sheets.

INSTRUCTIONS TO CANDIDATES

SECTION A – Product Design (60 marks)

Answer question 1.

SECTION B – Design and Drawing (40 marks)

Candidates can draw using instruments or they can use computers with **AutoCAD** software.
This section comprises **three** options, **A**, **B** and **C**.

Each option has **four** questions of **ten** marks each.

Choose **one** option **only** and answer **all** the questions in the chosen option.

If **AutoCAD** is used:

- (i) Use **A3** interface,
- (ii) All answers should be printed on **A3** size bond paper.

This specimen paper consists of 11 printed pages.

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SECTION A

(PRODUCT DESIGN) (60 marks)

Answer question 1 using the given guidelines.

- 1** **Figure 1** shows slices of hot toasted bread.

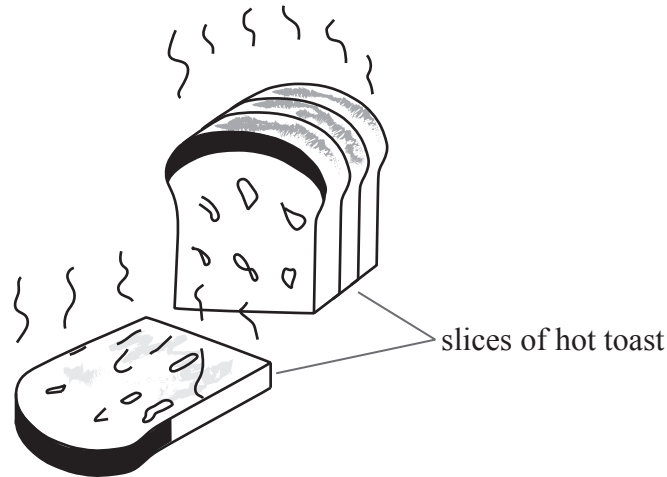


Fig. 1

Design a toast holder that can accommodate a number of hot slices of toasted bread.

- (a) List any **three** additional factors that can enhance the function of the toast holder. [3]

- (b) Using different illustrative techniques, sketch **three** different ideas of the toast holder. [18]

- (c)
 - (i) Evaluate each of the **three** ideas. [6]
 - (ii) Choose **one** idea from the ideas in (i). [1]
 - (iii) Justify your choice. [1]
 - (iv) Develop the chosen idea. [3]

- (d) Draw, using a method of choice, the completed toast holder. Include construction details and important dimensions. [14]

- (e)
 - (i) Suggest any **two** suitable specific materials for any **two** major components of the design. [2]
 - (ii) Give a reason for the choice of each material in (e)(i). [2]

- (f) Use sketches and notes to show **two** methods of joining any **two** components of the toast holder in (d). [6]

- (g) Outline a method that could be used to manufacture any **one** component of the toast holder. [4]

SECTION B (40 marks)

OPTION A (40 marks)

Choose **one** option only and answer **all** questions in the chosen option.

- 2 (a) **Figure 2(a)** shows a clothes' drying line attached to a concrete pillar. Connected to the concrete pillar is a machine which can tighten the clothes line when the wing nut is turned manually.

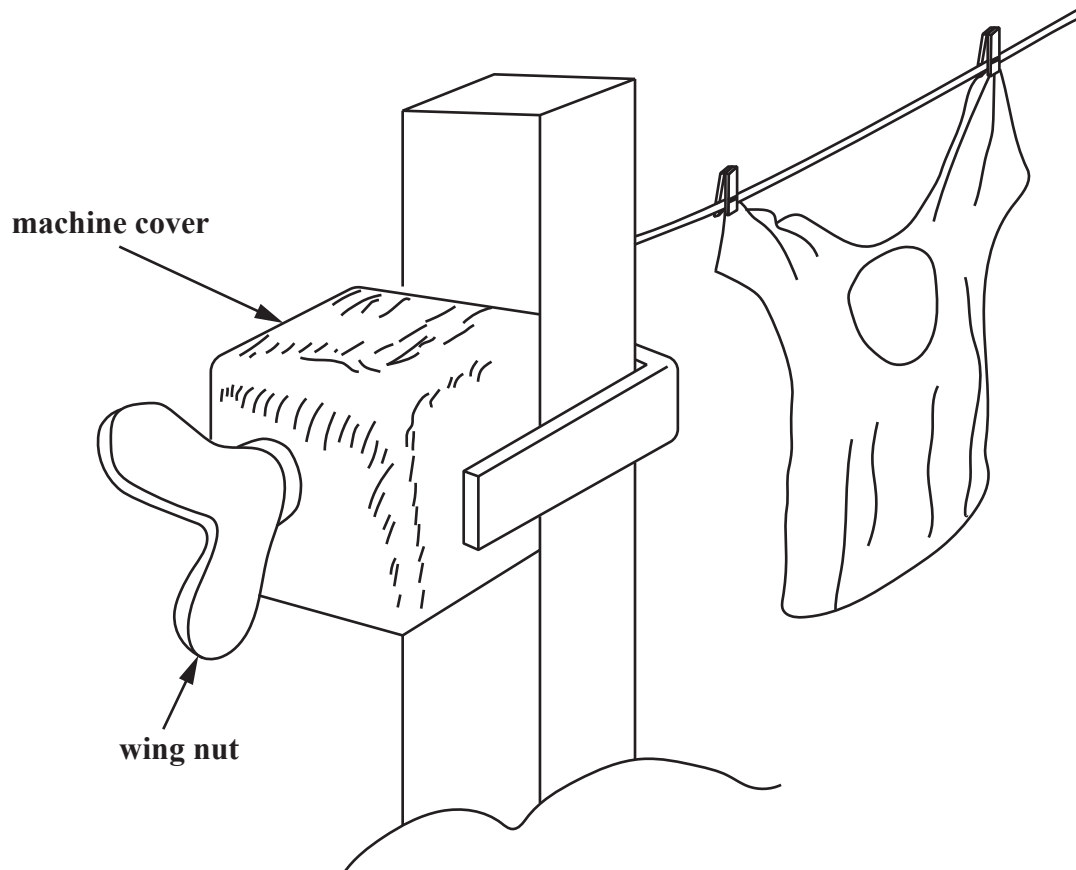


Fig. 2(a)

- (i) Using a drawing method of choice, show how a **worm gear system** can be used to tighten the clothes line. [7]
 - (ii) Calculate the pitch of the worm gear teeth given that the diameter of the wheel is **120** and velocity ratio is **48:1**. [3]
- (b) A newly established residential area needs electricity. You have been assigned to mobilise people to make contributions towards electrification of the area.

Draw a poster showing the importance of electrical energy. [10]

(c) **Figure 2(c)** shows electrical components on a circuit board.

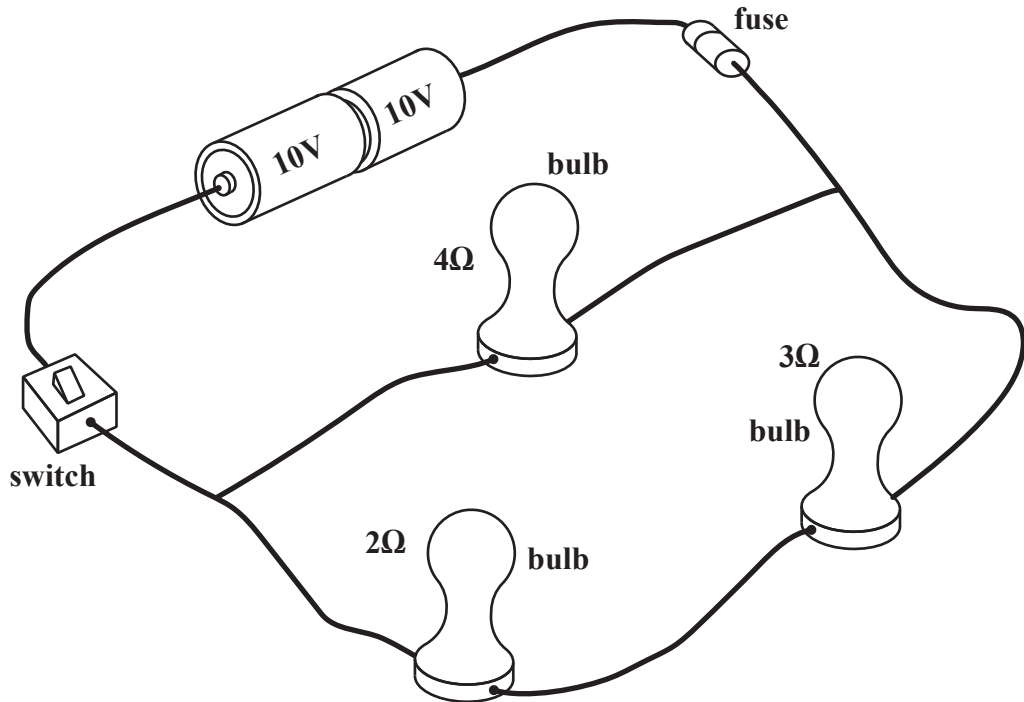


Fig. 2(c)

(i) Draw a schematic diagram of the circuit. [6]

(ii) Calculate the:

1. total resistance of the circuit, [2]

2. current across the fuse. [2]

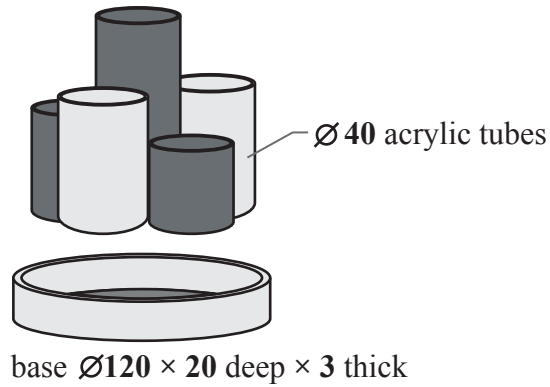
(d) Studies were carried out to establish type of fuels and forms of energy families in rural areas, growth points and plots were using for cooking. The data collected was tabulated as shown in the following table:

Family location	Firewood	Gas	Hydroelectricity	Solar	Charcoal
Rural areas	80	5	10	100	6
Growth points	10	130	5	200	20
Plots	8	8	70	20	3

Use a graph of your choice to illustrate the tabulated information. [10]

OPTION B (40 marks)

- 3 (a) **Figure 3 (a)** shows an exploded view of a desk tidy. The desk tidy consists of **five** differently coloured acrylic tubes and a base.

**Fig. 3(a)**

Heights of the acrylic tubes are **100, 70, 60, 55 and 40 mm** respectively.

Draw, using **First angle** or **Third angle projection** the following views of the assembled desk tidy:

- (i) front elevation,

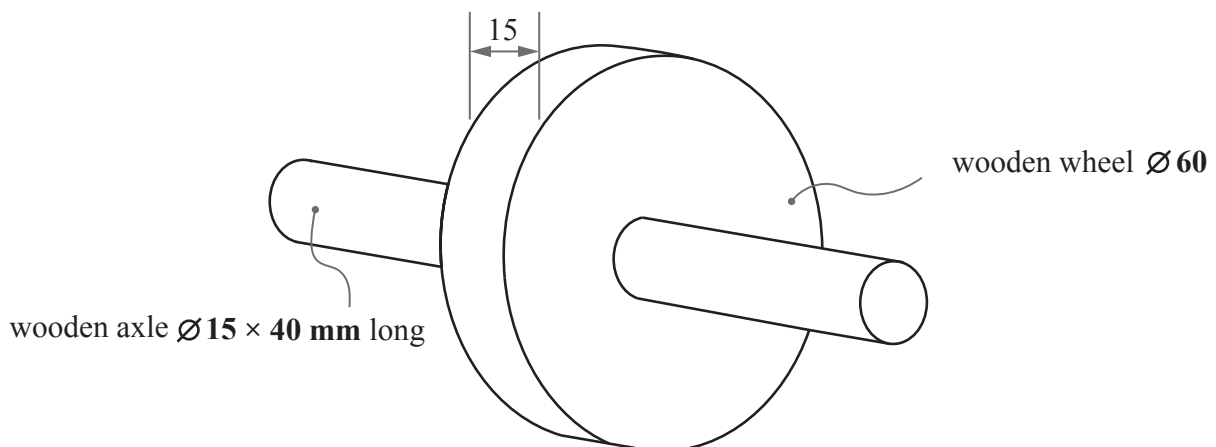
Use different colours or any rendering techniques to show the acrylic tubes and the base.

[5]

- (ii) plan.

[5]

- (b) **Figure 3(b)** shows a model of a wheel and axle for a wooden toy.

**Fig. 3(b)**

- (i) With the axle removed, construct the wheel to a scale of **full size**. [8]
- (ii) Apply a suitable rendering technique to show the material making up the wheel, in (i). [2]
- (c) **Figure 3(c)** shows an incomplete design of a desk that can be made in a school workshop.

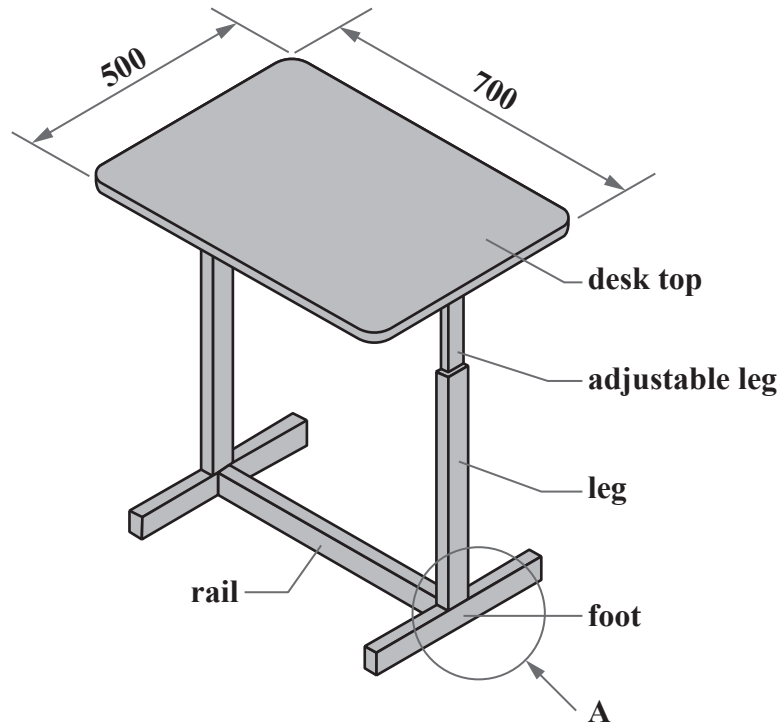


Fig. 3(c)

Materials list

leg, foot, rail	?
adjustable leg	mild steel tube 50 x 25
desk top	?

- (i) Suggest any suitable material for:
1. the desk top. [1]
 2. leg, foot and rail set. [1]
- (ii) Identify a suitable joining method for the joints at **A**. [1]
- (iii) Name the type of joints at **A**. [2]
- (iv) Draw to a scale of **full size**, an exploded view of the joint at **A**. [5]

- (d) **Figure 3(d)** shows the outline of a puzzle piece for a child's shapes game. The piece is made from a bright coloured acrylic material.

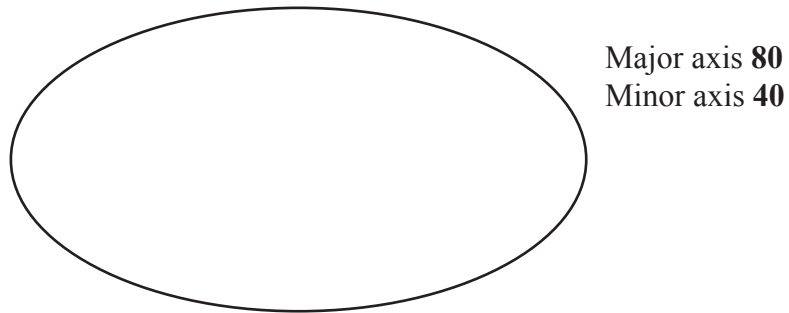


Fig. 3(d)

- (i) Draw the elliptical outline of the piece. [7]
- (ii) Render or colour the diagram in (i) using any **one** primary colour. [3]

OPTION C (40 marks)

- 4 (a) **Figure 4 (a)** shows some electrical components in a circuit.

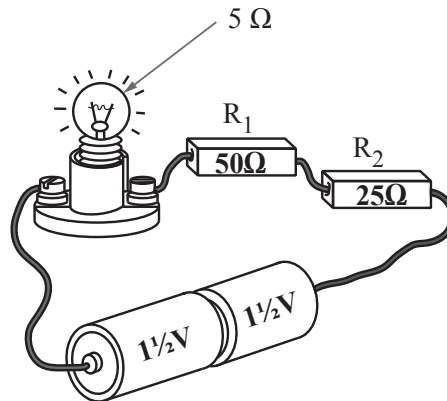


Fig. 4 (a)

- (i) Draw a schematic diagram for the given arrangement. [6]
- (ii) Calculate the total resistance in the circuit given that, resistor **R₁** is **50 Ω**, resistor **R₂** is **25 Ω** and bulb **5 Ω**. [2]
- (iv) What other component can be added to the circuit to ensure that the bulb glows only when needed. [2]

- (b) **Figure 4(b)** shows a **2 D** drawing.

Using prompts and commands, describe the steps of drawing **Fig. 4(b)**, using a drawing software of choice. [10]

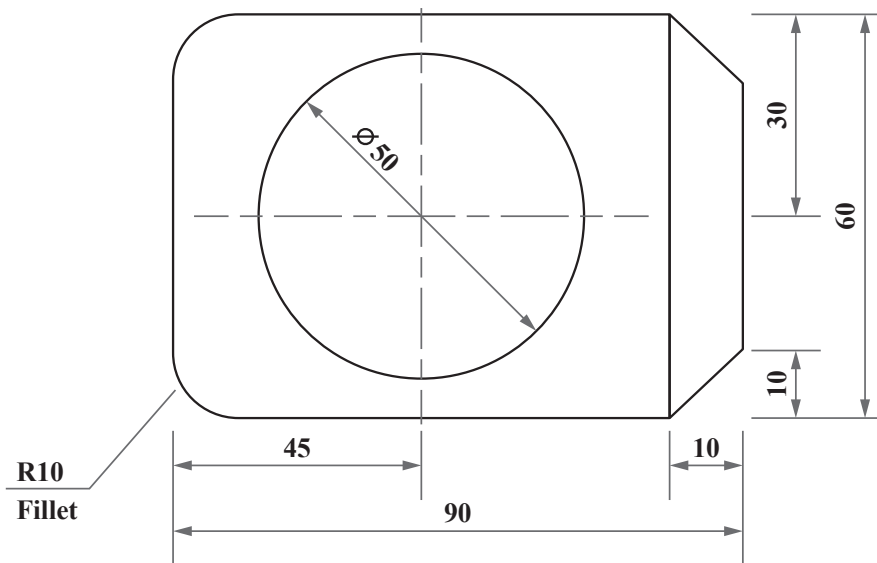


Fig. 4(b)

- (c) **Figure 4(c)** shows a design for a symmetrical bridge. The arc of the bridge is semi-elliptical, major axis is **10 m** and minor axis is **6 m**.

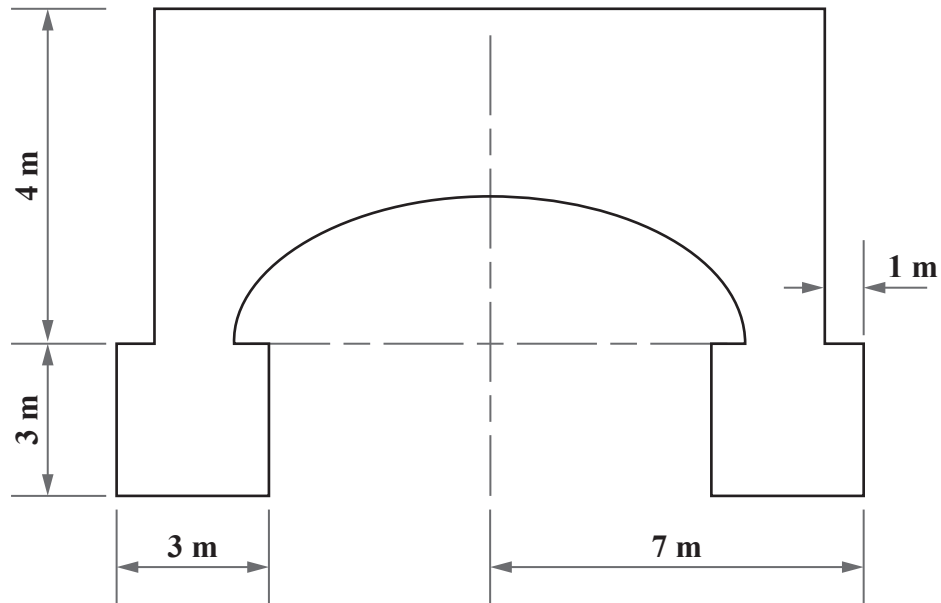


Fig. 4(c)

- (i) Draw, the given view to a scale of **1:100** using instruments. [6]
- (ii) Give any **two** advantages of using arches on bridges. [2]
- (iii) Identify any **two** materials suitable for bridge construction. [2]

- (d) **Figure 4(d)** shows a desk name display unit made of sheet metal.

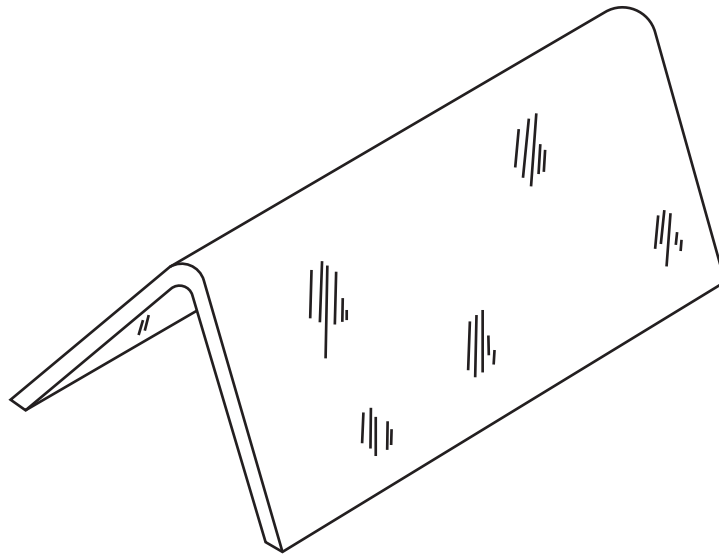


Fig. 4(d)

- (i) Draw a development of the given unit. Take measurements from the diagram. [5]
- (ii) Write the word '**HEAD**' on one of the developed faces. Use a lettering style of your choice. [2]
- (iii) Name any **two** tools that were used to fold the unit. [2]
- (v) State any **one** safety precaution observed when folding the unit. [1]